

作业6： 位移法求解正交各向异性直梁的弯曲

基本方程

平面应力问题前提条件:

对于厚度较小的直梁，外力平行于梁平面并沿厚度不变，即：

$\sigma_y = \tau_{xy} = \tau_{yz} = 0$

于是三维基本方程均简化：

- 几何方程：
$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_z = \frac{\partial w}{\partial z}, \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$
- 平衡方程（不计体力）：
$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0, \quad \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \sigma_z}{\partial z} = 0$$
- 本构关系（正交各向异性材料）：
$$\begin{aligned} \sigma_x &= c_{11}\varepsilon_x + c_{13}\varepsilon_z \\ \sigma_z &= c_{13}\varepsilon_x + c_{33}\varepsilon_z \\ \tau_{zx} &= c_{55}\gamma_{zx} \end{aligned}$$

其中  $c_{ij} = c_{ij}(x, z)$ 。

控制方程：

从基本方程中推导仅有位移变量的控制方程

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In [1]: import sympy as sp

# 定义符号变量
x, z = sp.symbols('x z')
u, w = sp.Function('u')(x, z), sp.Function('w')(x, z)
c11, c13, c33, c55 = sp.Function('c11')(x, z), sp.Function('c13')(x, z), sp.Function('c33')(x, z), sp.Function('c55')(x, z)

In [2]: print("=== 平面应力问题位移法控制方程推导 ===\n")

=== 平面应力问题位移法控制方程推导 ===

In [3]: print("1. 几何方程：")
epsilon_x = sp.diff(u, x)
epsilon_z = sp.diff(w, z)
gamma_zx = sp.diff(w, x) + sp.diff(u, z)

print(f"ε_x = ∂u/∂x = {epsilon_x}")
print(f"ε_z = ∂w/∂z = {epsilon_z}")
print(f"γ_zx = ∂w/∂x + ∂u/∂z = {gamma_zx}\n")

1. 几何方程：
ε_x = ∂u/∂x = Derivative(u(x, z), x)
ε_z = ∂w/∂z = Derivative(w(x, z), z)
γ_zx = ∂w/∂x + ∂u/∂z = Derivative(u(x, z), z) + Derivative(w(x, z), x)

In [4]: print("2. 本构关系（应力用位移表示）：")
sigma_x = c11 * epsilon_x + c13 * epsilon_z
sigma_z = c13 * epsilon_x + c33 * epsilon_z
tau_zx = c55 * gamma_zx

print(f"σ_x = c11·ε_x + c13·ε_z = {sigma_x}")
print(f"σ_z = c13·ε_x + c33·ε_z = {sigma_z}")
print(f"τ_zx = c55·γ_zx = {tau_zx}\n")

2. 本构关系（应力用位移表示）：
σ_x = c11·ε_x + c13·ε_z = c11(x, z)*Derivative(u(x, z), x) + c13(x, z)*Derivative(w(x, z), z)
σ_z = c13·ε_x + c33·ε_z = c13(x, z)*Derivative(u(x, z), x) + c33(x, z)*Derivative(w(x, z), z)
τ_zx = c55·γ_zx = (Derivative(u(x, z), z) + Derivative(w(x, z), x))*c55(x, z)

In [5]: print("3. 平衡方程：")
print("∂σ_x/∂x + ∂τ_zx/∂z = 0")
print("∂τ_zx/∂x + ∂σ_z/∂z = 0\n")

print("4. 将应力代入平衡方程：")
print("第一个平衡方程：")
eq1 = sp.diff(sigma_x, x) + sp.diff(tau_zx, z)
print(f"∂σ_x/∂x + ∂τ_zx/∂z = {eq1}")

print("\n第二个平衡方程：")
eq2 = sp.diff(tau_zx, x) + sp.diff(sigma_z, z)
print(f"∂τ_zx/∂x + ∂σ_z/∂z = {eq2}\n")

3. 平衡方程：
∂σ_x/∂x + ∂τ_zx/∂z = 0
∂τ_zx/∂x + ∂σ_z/∂z = 0

4. 将应力代入平衡方程：
第一个平衡方程：
∂σ_x/∂x + ∂τ_zx/∂z = (Derivative(u(x, z), z) + Derivative(w(x, z), x))*Derivative(c55(x, z), z) + (Derivative(u(x, z), (z, 2)) + Derivative(w(x, z), x, z))*c55(x, z) + c11(x, z)*Derivative(u(x, z), (x, 2)) + c13(x, z)*Derivative(w(x, z), x, z) + Derivative(c11(x, z), x)*Derivative(u(x, z), x) + Derivative(c13(x, z), x)*Derivative(w(x, z), z)

第二个平衡方程：
∂τ_zx/∂x + ∂σ_z/∂z = (Derivative(u(x, z), z) + Derivative(w(x, z), x))*Derivative(c55(x, z), x) + (Derivative(w(x, z), (x, 2)) + Derivative(u(x, z), x, z))*c55(x, z) + c13(x, z)*Derivative(u(x, z), x, z) + c33(x, z)*Derivative(w(x, z), (z, 2)) + Derivative(c13(x, z), z)*Derivative(u(x, z), x) + Derivative(c33(x, z), z)*Derivative(w(x, z), z)

In [6]: print("5. 展开并整理控制方程：")
print("第一个控制方程（u方向）：")
# 展开导数
eq1_expanded = sp.expand(eq1)
print(f"展开后：{eq1_expanded}")

# 按位移导数分类整理
u_xx_term = c11 * sp.diff(u, x, x)
u_zz_term = c55 * sp.diff(u, z, z)
u_xz_term = (sp.diff(c11, x) + sp.diff(c55, z)) * sp.diff(u, x) + (c13 + c55) * sp.diff(u, x, z)
w_xz_term = (c13 + c55) * sp.diff(w, x, z) + sp.diff(c13, x) * sp.diff(w, z) + sp.diff(c55, z) * sp.diff(w, x)

eq1_final = u_xx_term + u_zz_term + u_xz_term + w_xz_term
print(f"整理后：{eq1_final} = 0\n")

print("第二个控制方程（w方向）：")
eq2_expanded = sp.expand(eq2)
print(f"展开后：{eq2_expanded}")

# 按位移导数分类整理
w_xx_term = c55 * sp.diff(w, x, x)
w_zz_term = c33 * sp.diff(w, z, z)
w_xz_term = (sp.diff(c55, x) + sp.diff(c33, z)) * sp.diff(w, z) + (c13 + c55) * sp.diff(w, x, z)
u_xz_term = (c13 + c55) * sp.diff(u, x, z) + sp.diff(c55, x) * sp.diff(u, z) + sp.diff(c13, z) * sp.diff(u, x)
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eq2_final = w_xx_term + w_zz_term + w_xz_term + u_xz_term
print(f"整理后：{eq2_final} = 0\n")
```

5. 展开并整理控制方程：  
第一个控制方程（u方向）：  
展开后：c11(x, z)\*Derivative(u(x, z), (x, 2)) + c13(x, z)\*Derivative(w(x, z), x, z) + c55(x, z)\*Derivative(u(x, z), (z, 2)) + c55(x, z)\*Derivative(w(x, z), x, z) + Derivative(c11(x, z), x)\*Derivative(u(x, z), x) + Derivative(c13(x, z), x)\*Derivative(w(x, z), z) + Derivative(c55(x, z), z)\*Derivative(u(x, z), z) + Derivative(c55(x, z), z)\*Derivative(w(x, z), x)  
整理后：(c13(x, z) + c55(x, z))\*Derivative(u(x, z), x, z) + (c13(x, z) + c55(x, z))\*Derivative(w(x, z), x, z) + (Derivative(c11(x, z), x) + Derivative(c55(x, z), z))\*Derivative(u(x, z), x) + c11(x, z)\*Derivative(u(x, z), (x, 2)) + c55(x, z)\*Derivative(u(x, z), (z, 2)) + Derivative(c13(x, z), x)\*Derivative(w(x, z), z) + Derivative(c55(x, z), z)\*Derivative(w(x, z), x) = 0  
  
第二个控制方程（w方向）：  
展开后：c13(x, z)\*Derivative(u(x, z), x, z) + c33(x, z)\*Derivative(w(x, z), (z, 2)) + c55(x, z)\*Derivative(w(x, z), (x, 2)) + c55(x, z)\*Derivative(u(x, z), x, z) + Derivative(c13(x, z), z)\*Derivative(u(x, z), x) + Derivative(c33(x, z), z)\*Derivative(w(x, z), z) + Derivative(c55(x, z), x)\*Derivative(u(x, z), z) + Derivative(c55(x, z), x)\*Derivative(w(x, z), x)  
整理后：(c13(x, z) + c55(x, z))\*Derivative(u(x, z), x, z) + (c13(x, z) + c55(x, z))\*Derivative(w(x, z), x, z) + (Derivative(c33(x, z), z) + Derivative(c55(x, z), x))\*Derivative(w(x, z), z) + c33(x, z)\*Derivative(w(x, z), (z, 2)) + c55(x, z)\*Derivative(w(x, z), (x, 2)) + Derivative(c13(x, z), z)\*Derivative(u(x, z), x) + Derivative(c55(x, z), x)\*Derivative(u(x, z), z) = 0

```
In [7]: print("6. 最终位移法控制方程组：")
print("对于正交各向异性材料，平面应力问题的位移法控制方程为：")
print(f"c11·∂²u/∂x² + c55·∂²u/∂z² + (∂c11/∂x + ∂c55/∂z)·∂u/∂x + (c13 + c55)·∂²u/∂x∂z")
print(f" + (c13 + c55)·∂²w/∂x∂z + ∂c13/∂x·∂w/∂z + ∂c55/∂z·∂w/∂x = 0")
print()
print(f"c55·∂²w/∂x² + c33·∂²w/∂z² + (∂c55/∂x + ∂c33/∂z)·∂w/∂z + (c13 + c55)·∂²w/∂x∂z")
print(f" + (c13 + c55)·∂²u/∂x∂z + ∂c55/∂x·∂u/∂z + ∂c13/∂z·∂u/∂x = 0")

print("\n7. 特殊情况：均匀材料（材料常数与位置无关）")
print("当 c11, c13, c33, c55 为常数时，控制方程简化为：")
print("c11·∂²u/∂x² + c55·∂²u/∂z² + (c13 + c55)·∂²w/∂x∂z = 0")
print("c55·∂²w/∂x² + c33·∂²w/∂z² + (c13 + c55)·∂²u/∂x∂z = 0")
```

6. 最终位移法控制方程组：  
对于正交各向异性材料，平面应力问题的位移法控制方程为：  
c11·∂²u/∂x² + c55·∂²u/∂z² + (∂c11/∂x + ∂c55/∂z)·∂u/∂x + (c13 + c55)·∂²u/∂x∂z + (c13 + c55)·∂²w/∂x∂z + ∂c13/∂x·∂w/∂z + ∂c55/∂z·∂w/∂x = 0  
  
c55·∂²w/∂x² + c33·∂²w/∂z² + (∂c55/∂x + ∂c33/∂z)·∂w/∂z + (c13 + c55)·∂²w/∂x∂z + (c13 + c55)·∂²u/∂x∂z + ∂c55/∂x·∂u/∂z + ∂c13/∂z·∂u/∂x = 0

7. 特殊情况：均匀材料（材料常数与位置无关）  
当 c11, c13, c33, c55 为常数时，控制方程简化为：  
c11·∂²u/∂x² + c55·∂²u/∂z² + (c13 + c55)·∂²w/∂x∂z = 0  
c55·∂²w/∂x² + c33·∂²w/∂z² + (c13 + c55)·∂²u/∂x∂z = 0

```
In [8]: import sympy as sp
from IPython.display import display, Math, Latex
sp.init_printing()

# 定义符号变量
x, z = sp.symbols('x z')
u, w = sp.Function('u')(x, z), sp.Function('w')(x, z)
c11, c13, c33, c55 = sp.Function('c_{11}')(x, z), sp.Function('c_{13}')(x, z), sp.Function('c_{33}')(x, z), sp.Function('c_{55}')(x, z)

print("=== 平面应力问题位移法控制方程推导 ===\n")

display(Math(r"\text{1. 几何方程: }"))
epsilon_x = sp.diff(u, x)
epsilon_z = sp.diff(w, z)
gamma_zx = sp.diff(w, x) + sp.diff(u, z)

display(Math(r"\varepsilon_x = \frac{\partial u}{\partial x}"))
display(Math(r"\varepsilon_z = \frac{\partial w}{\partial z}"))
display(Math(r"\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}"))

display(Math(r"\text{2. 本构关系: }"))
sigma_x = c11 * epsilon_x + c13 * epsilon_z
sigma_z = c13 * epsilon_x + c33 * epsilon_z
tau_zx = c55 * gamma_zx

display(Math(r"\text{将几何方程带入本构关系，得到位移表示的应力表达式: }"))
display(Math(r"\sigma_x = c_{11}\varepsilon_x + c_{13}\varepsilon_z"))
display(Math(r"\sigma_z = c_{13}\varepsilon_x + c_{33}\varepsilon_z"))
display(Math(r"\tau_{zx} = c_{55}\gamma_{zx}"))

=== 平面应力问题位移法控制方程推导 ===
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1.几何方程：

$$\varepsilon_x = \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}u(x,z)$$
$$\varepsilon_z = \frac{\partial w}{\partial z} = \frac{\partial}{\partial z}w(x,z)$$
$$\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = \frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)$$

2. 本构关系：

将几何方程带入本构关系，得到位移表示的应力表达式：

$$\sigma_x = c_{11}\varepsilon_x + c_{13}\varepsilon_z = c_{11}(x,z)\frac{\partial}{\partial x}u(x,z) + c_{13}(x,z)\frac{\partial}{\partial z}w(x,z)$$
$$\sigma_z = c_{13}\varepsilon_x + c_{33}\varepsilon_z = c_{13}(x,z)\frac{\partial}{\partial x}u(x,z) + c_{33}(x,z)\frac{\partial}{\partial z}w(x,z)$$
$$\tau_{zx} = c_{55}\gamma_{zx} = \left(\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)\right)c_{55}(x,z)$$

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In [9]: display(Math(r"\text{3. 平衡方程: }"))
display(Math(r"\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0"))
display(Math(r"\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \sigma_z}{\partial z} = 0"))

display(Math(r"\text{将应力的位移表达式代入平衡方程: }"))
eq1 = sp.diff(sigma_x, x) + sp.diff(tau_zx, z)
eq2 = sp.diff(tau_zx, x) + sp.diff(sigma_z, z)
display(Math(r"\text{位移法控制方程: }"))
display(Math(r"\text{方程1: }" + sp.latex(sp.Eq(eq1, 0)))))
display(Math(r"\text{方程2: }" + sp.latex(sp.Eq(eq2, 0)))))

display(Math(r"\text{展开并整理控制方程: }"))
display(Math(r"\text{第一个控制方程（u方向）: }"))
eq1_expanded = sp.expand(eq1)
display(Math(r"\text{展开后: }" + sp.latex(eq1_expanded)))
u_xx_term = c11 * sp.diff(u, x, x)
u_zz_term = c55 * sp.diff(u, z, z)
u_xz_term = (sp.diff(c11, x) + sp.diff(c55, z)) * sp.diff(u, x) + (c13 + c55) * sp.diff(u, x, z)
w_xz_term = (c13 + c55) * sp.diff(w, x, z) + sp.diff(c13, x) * sp.diff(w, z) + sp.diff(c55, z) * sp.diff(w, x)
eq1_final = u_xx_term + u_zz_term + u_xz_term + w_xz_term
display(Math(r"\text{整理后: }" + sp.latex(sp.Eq(eq1_final, 0)))))
display(Math(r"\text{第二个控制方程（w方向）: }"))
eq2_expanded = sp.expand(eq2)
display(Math(r"\text{展开后: }" + sp.latex(eq2_expanded)))
w_xx_term = c55 * sp.diff(w, x, x)
w_zz_term = c33 * sp.diff(w, z, z)
w_xz_term = (sp.diff(c55, x) + sp.diff(c33, z)) * sp.diff(w, z) + (c13 + c55) * sp.diff(w, x, z)
u_xz_term = (c13 + c55) * sp.diff(u, x, z) + sp.diff(c55, x) * sp.diff(u, z) + sp.diff(c13, z) * sp.diff(u, x)
eq2_final = w_xx_term + w_zz_term + w_xz_term + u_xz_term
display(Math(r"\text{整理后: }" + sp.latex(sp.Eq(eq2_final, 0)))))
display(Math(r"\text{最终位移法控制方程组: }"))
display(Math(r"\text{对于正交各向异性材料，平面应力问题的位移法控制方程为: }"))
display(Math(r"c_{11}\frac{\partial^2 u}{\partial x^2} + c_{55}\frac{\partial^2 u}{\partial z^2} + \left(\frac{\partial c_{11}}{\partial x} + \frac{\partial c_{55}}{\partial z}\right)\frac{\partial u}{\partial x} + \left(\frac{\partial c_{13}}{\partial x} + \frac{\partial c_{55}}{\partial z}\right)\frac{\partial w}{\partial z} + \frac{\partial c_{13}}{\partial x}\frac{\partial w}{\partial x} + \frac{\partial c_{55}}{\partial z}\frac{\partial u}{\partial z} + c_{13}\frac{\partial^2 u}{\partial x\partial z} + c_{33}\frac{\partial^2 w}{\partial z^2} + \left(\frac{\partial c_{55}}{\partial x} + \frac{\partial c_{33}}{\partial z}\right)\frac{\partial w}{\partial z} + (c_{13} + c_{55})\frac{\partial^2 w}{\partial x\partial z} + \frac{\partial c_{55}}{\partial x}\frac{\partial u}{\partial z} + \frac{\partial c_{13}}{\partial z}\frac{\partial u}{\partial x} = 0"))
```

3. 平衡方程：

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \sigma_z}{\partial z} = 0$$

将应力的位移表达代入平衡方程：

位移法控制方程：

方程1:  $\left(\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)\right)\frac{\partial}{\partial z}c_{55}(x,z) + \left(\frac{\partial^2}{\partial z^2}u(x,z) + \frac{\partial^2}{\partial z\partial x}w(x,z)\right)c_{55}(x,z) + c_{11}(x,z)\frac{\partial^2}{\partial x^2}u(x,z) + c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}w(x,z) + \frac{\partial}{\partial x}c_{11}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial x}c_{13}(x,z)\frac{\partial}{\partial z}w(x,z) = 0$

方程2:  $\left(\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)\right)\frac{\partial}{\partial x}c_{55}(x,z) + \left(\frac{\partial^2}{\partial x^2}w(x,z) + \frac{\partial^2}{\partial z\partial x}u(x,z)\right)c_{55}(x,z) + c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}u(x,z) + c_{33}(x,z)\frac{\partial^2}{\partial z^2}w(x,z) + \frac{\partial}{\partial z}c_{13}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial z}c_{33}(x,z)\frac{\partial}{\partial z}w(x,z) = 0$

展开并整理控制方程：

第一个控制方程（u方向）：

展开后:  $c_{11}(x,z)\frac{\partial^2}{\partial x^2}u(x,z) + c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}w(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial z^2}u(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial z\partial x}w(x,z) + \frac{\partial}{\partial x}c_{11}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial x}c_{13}(x,z)\frac{\partial}{\partial z}w(x,z) + \frac{\partial}{\partial z}c_{55}(x,z)\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial z}c_{55}(x,z)\frac{\partial}{\partial x}w(x,z)$

整理后:  $(c_{13}(x,z) + c_{55}(x,z))\frac{\partial^2}{\partial z\partial x}u(x,z) + (c_{13}(x,z) + c_{55}(x,z))\frac{\partial^2}{\partial z\partial x}w(x,z) + \left(\frac{\partial}{\partial x}c_{11}(x,z) + \frac{\partial}{\partial z}c_{55}(x,z)\right)\frac{\partial}{\partial x}u(x,z) + c_{11}(x,z)\frac{\partial^2}{\partial x^2}u(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial z^2}u(x,z) + \frac{\partial}{\partial x}c_{13}(x,z)\frac{\partial}{\partial z}w(x,z)$   
 $+ \frac{\partial}{\partial z}c_{55}(x,z)\frac{\partial}{\partial x}w(x,z) = 0$

第二个控制方程（w方向）：

展开后:  $c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}u(x,z) + c_{33}(x,z)\frac{\partial^2}{\partial z^2}w(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial x^2}w(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial z\partial x}u(x,z) + \frac{\partial}{\partial z}c_{13}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial z}c_{33}(x,z)\frac{\partial}{\partial z}w(x,z) + \frac{\partial}{\partial x}c_{55}(x,z)\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}c_{55}(x,z)\frac{\partial}{\partial x}w(x,z)$

整理后:  $(c_{13}(x,z) + c_{55}(x,z))\frac{\partial^2}{\partial z\partial x}u(x,z) + (c_{13}(x,z) + c_{55}(x,z))\frac{\partial^2}{\partial z\partial x}w(x,z) + \left(\frac{\partial}{\partial z}c_{33}(x,z) + \frac{\partial}{\partial x}c_{55}(x,z)\right)\frac{\partial}{\partial z}w(x,z) + c_{33}(x,z)\frac{\partial^2}{\partial z^2}w(x,z) + c_{55}(x,z)\frac{\partial^2}{\partial x^2}w(x,z) + \frac{\partial}{\partial z}c_{13}(x,z)\frac{\partial}{\partial x}u(x,z)$   
 $+ \frac{\partial}{\partial x}c_{55}(x,z)\frac{\partial}{\partial z}u(x,z) = 0$

最终位移法控制方程组：

对于正交各向异性材料，平面应力问题的位移法控制方程为：

$$c_{11}\frac{\partial^2 u}{\partial x^2} + c_{55}\frac{\partial^2 u}{\partial z^2} + \left(\frac{\partial c_{11}}{\partial x} + \frac{\partial c_{55}}{\partial z}\right)\frac{\partial u}{\partial x} + (c_{13} + c_{55})\frac{\partial^2 u}{\partial x\partial z}$$
$$+ (c_{13} + c_{55})\frac{\partial^2 w}{\partial x\partial z} + \frac{\partial c_{13}}{\partial x}\frac{\partial w}{\partial z} + \frac{\partial c_{55}}{\partial z}\frac{\partial w}{\partial x} = 0$$
$$c_{55}\frac{\partial^2 w}{\partial x^2} + c_{33}\frac{\partial^2 w}{\partial z^2} + \left(\frac{\partial c_{55}}{\partial x} + \frac{\partial c_{33}}{\partial z}\right)\frac{\partial w}{\partial z} + (c_{13} + c_{55})\frac{\partial^2 w}{\partial x\partial z}$$
$$+ (c_{13} + c_{55})\frac{\partial^2 u}{\partial x\partial z} + \frac{\partial c_{55}}{\partial x}\frac{\partial u}{\partial z} + \frac{\partial c_{13}}{\partial z}\frac{\partial u}{\partial x} = 0$$

```
In [10]: display(Math(r"\text{位移法控制方程: }"))
eq1 = sp.diff(sigma_x, x) + sp.diff(tau_zx, z)
eq2 = sp.diff(tau_zx, x) + sp.diff(sigma_z, z)

display(Math(r"\text{方程1: }" + sp.latex(sp.Eq(eq1, 0))))
display(Math(r"\text{方程2: }" + sp.latex(sp.Eq(eq2, 0))))
```

位移法控制方程：

方程1:  $\left(\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)\right)\frac{\partial}{\partial z}c_{55}(x,z) + \left(\frac{\partial^2}{\partial z^2}u(x,z) + \frac{\partial^2}{\partial z\partial x}w(x,z)\right)c_{55}(x,z) + c_{11}(x,z)\frac{\partial^2}{\partial x^2}u(x,z) + c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}w(x,z) + \frac{\partial}{\partial x}c_{11}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial x}c_{13}(x,z)\frac{\partial}{\partial z}w(x,z) = 0$

方程2:  $\left(\frac{\partial}{\partial z}u(x,z) + \frac{\partial}{\partial x}w(x,z)\right)\frac{\partial}{\partial x}c_{55}(x,z) + \left(\frac{\partial^2}{\partial x^2}w(x,z) + \frac{\partial^2}{\partial z\partial x}u(x,z)\right)c_{55}(x,z) + c_{13}(x,z)\frac{\partial^2}{\partial z\partial x}u(x,z) + c_{33}(x,z)\frac{\partial^2}{\partial z^2}w(x,z) + \frac{\partial}{\partial z}c_{13}(x,z)\frac{\partial}{\partial x}u(x,z) + \frac{\partial}{\partial z}c_{33}(x,z)\frac{\partial}{\partial z}w(x,z) = 0$